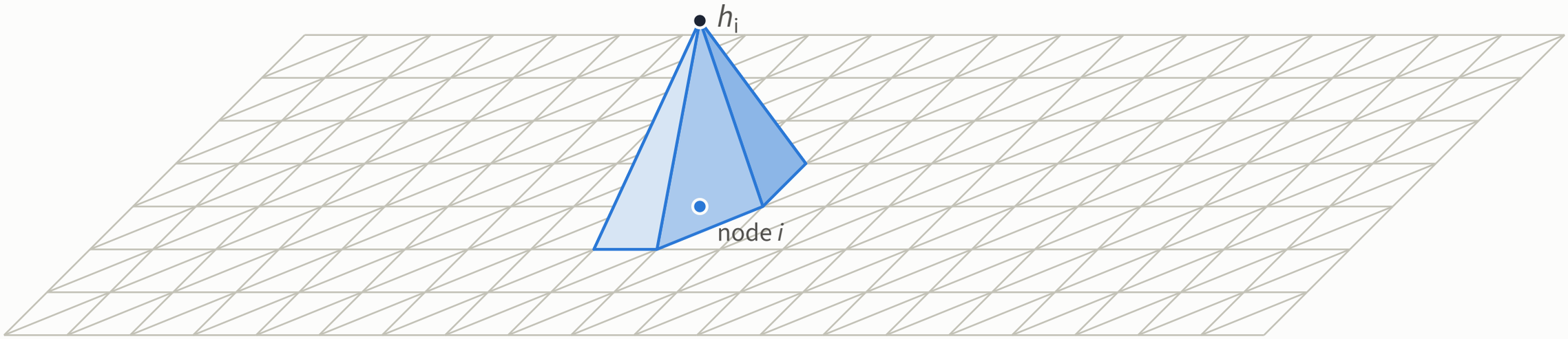
The finite element method

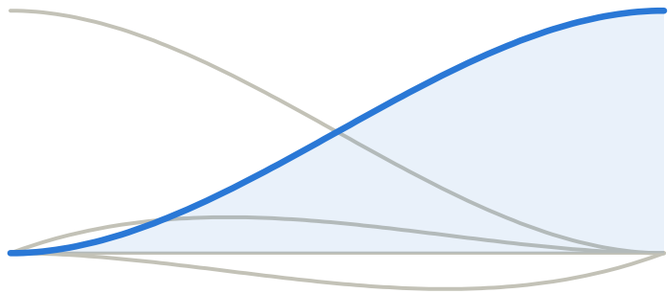
Local basis functions, sparse matrices and the conjugate gradient method



Global or local basis functions

The choice of $h_1, \ldots, h_N$ decides how well u_h approximates u and what the matrix K looks like.

Global basis functions

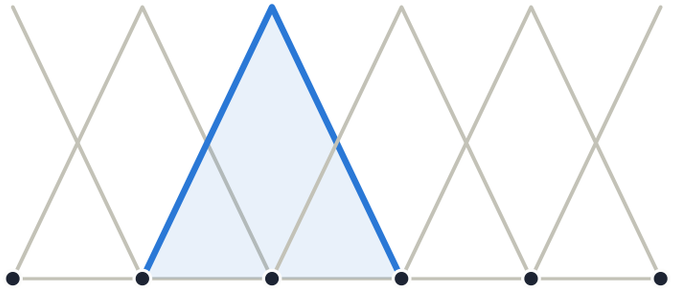




K

nonzero on the whole domain, e.g. the cubic Hermite polynomials of lecture 1. Every $K_{ji} \neq 0$: K is **dense**.

Local basis functions





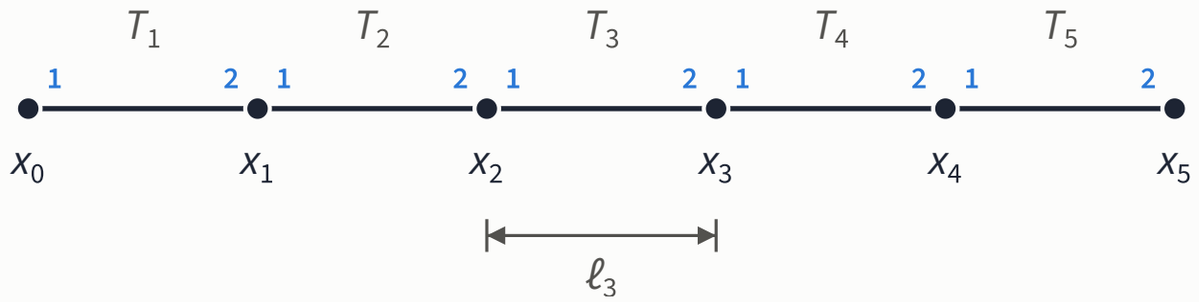
K

nonzero only on a small part of the domain. $a(h_i, h_j) = 0$ if the supports do not overlap: K is **sparse**.

The **finite element method** (FEM) is the Galerkin method with local basis functions.

The mesh and the coincidence table

The domain is approximated by a **mesh**, $\Omega \approx \Omega_h = \bigcup_e T_e$: a union of **elements** T_e (intervals, triangles, tetrahedra) that meet at their vertices, the **nodes**. On each element u_h is a polynomial of low degree.



In 1D the nodes $x_0 < x_1 < \cdots < x_N$ divide the interval into the elements $T_e = [x_{e-1}, x_e]$ of length $\ell_e = x_e - x_{e-1}$. Each element numbers its two nodes locally, 1 (left) and 2 (right).

Coincidence table

the global numbers of the local nodes of every element

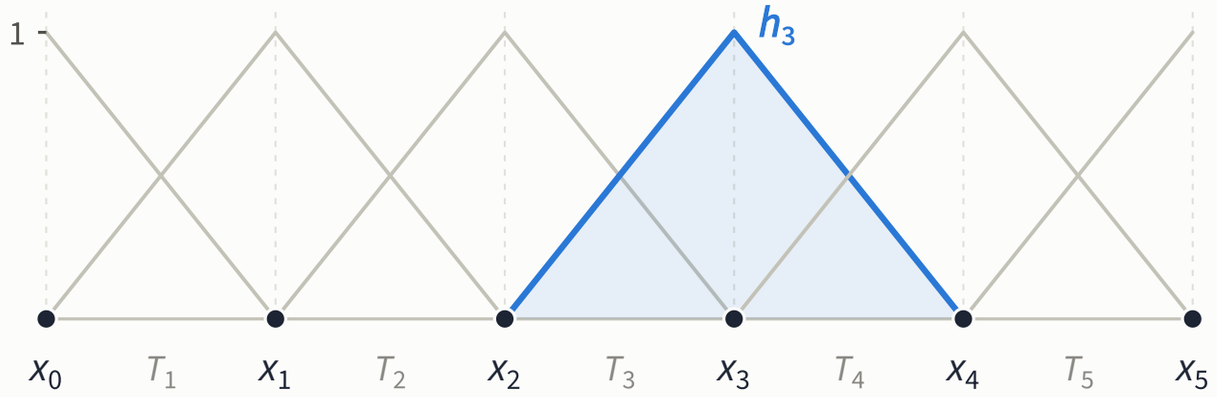
element	local node 1	local node 2
T_1	0	1
T_2	1	2
$\vdots$	$\vdots$	$\vdots$
T_N	$N-1$	N

In FEniCS: `mesh.cells()`

Hat functions in 1D

The hat function of node i:

$$h_i(x) = \begin{cases} (x - x_{i-1})/\ell_i & \text{for } x_{i-1} \le x \le x_i, \\ (x_{i+1} - x)/\ell_{i+1} & \text{for } x_i \le x \le x_{i+1}, \\ 0 & \text{otherwise.} \end{cases}$$



- $h_i(x_j) = \delta_{ij}$, so $U_i = u_h(x_i)$: u_h is the piecewise linear interpolation of the nodal values.
- On T_e only the hat functions of its two nodes are nonzero. In local numbering, h_α is the hat function of the local node α: on T_e, $h_1 = (x_e - x)/\ell_e$ and $h_2 = (x - x_{e-1})/\ell_e$.

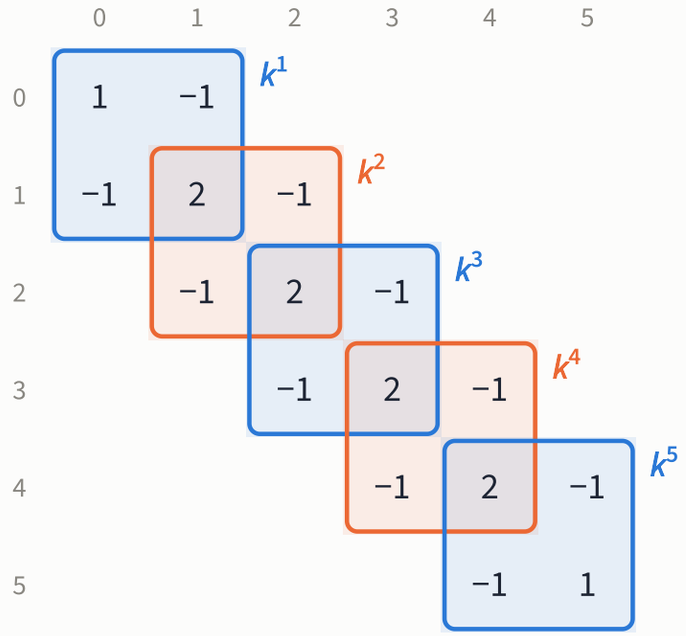
Element matrices and assembly

For $a(u,v) = \int u'\,v'\,dx$ and constant f, on the element T_e:

$$k^e = \left(\int_{T_e} h'_\alpha\, h'_\beta\, dx\right)_{\alpha\beta} = \frac{1}{\ell_e}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

$$f^e = \left(\int_{T_e} f\, h_\alpha\, dx\right)_{\alpha} = \frac{f\,\ell_e}{2}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Assembly. K is the sum of the element matrices. The coincidence table says where: $k^e_{\alpha\beta}$ is added to $K_{e-2+\alpha,\ e-2+\beta}$ and f^e_α to $F_{e-2+\alpha}$. Node i shares an element only with $i \pm 1$, so K is **tridiagonal**.



$h\,K$ for 5 elements of length h

The interior rows are the finite difference approximation $(-U_{i-1} + 2U_i - U_{i+1})/h^2$ of $-u''$, multiplied by h. The Dirichlet conditions replace the rows of the boundary nodes.

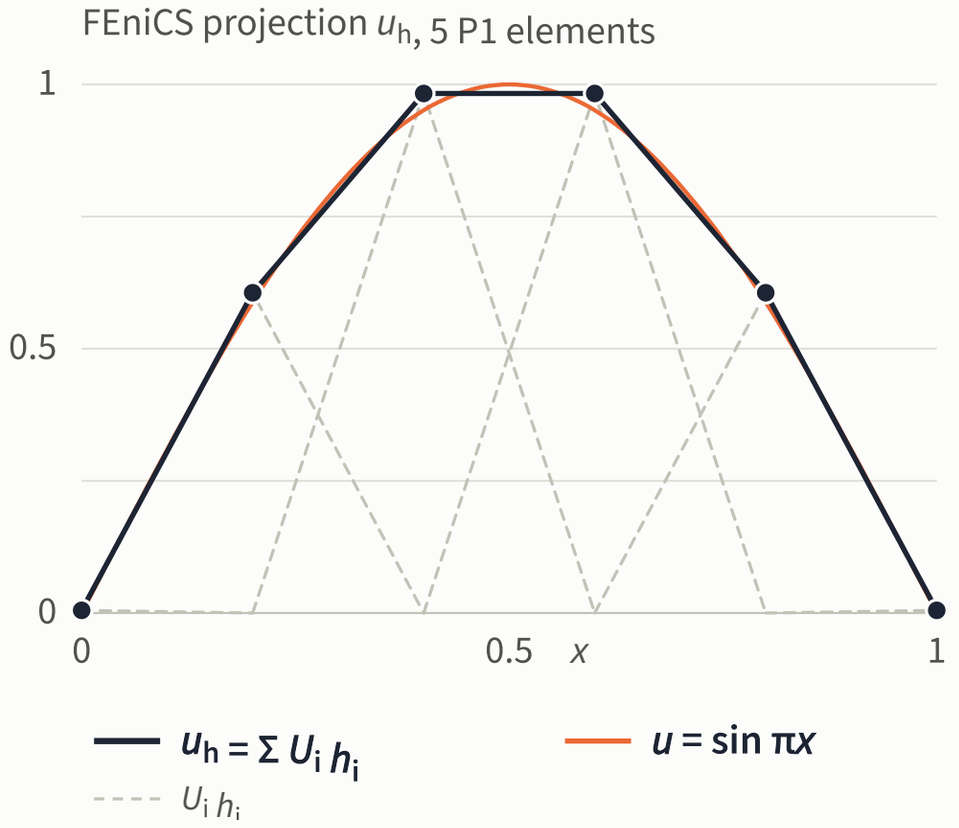
Hat functions and a projection

The Galerkin method with $a(u,v) = \int u\,v\,dx$ and $L(v) = \int \sin(\pi x)\,v\,dx$ gives $M\,U = F$, with the **mass matrix** $M_{ji} = \int h_i\,h_j\,dx$.

```
mesh = IntervalMesh(5, 0.0, 1.0)
V    = FunctionSpace(mesh, 'Lagrange', 1)

u, v = TrialFunction(V), TestFunction(V)
u_ex = Expression('sin(pi*x[0])', degree=4)
M    = assemble(u*v*dx)       # mass matrix
F    = assemble(u_ex*v*dx)    # load vector
uh   = Function(V)
solve(M, uh.vector(), F)

K = assemble(inner(grad(u), grad(v))*dx)
print(K.array())              # tridiagonal, 1/h = 5
```



Unlike the interpolation, u_h does not take the nodal values of u: $U_i = 0.983$ at $x = 0.4$, but $\sin(0.4\pi) = 0.951$. M is assembled from $m^e = \frac{\ell_e}{6}\begin{pmatrix}2 & 1\\ 1 & 2\end{pmatrix}$.

The linear triangle (CST)

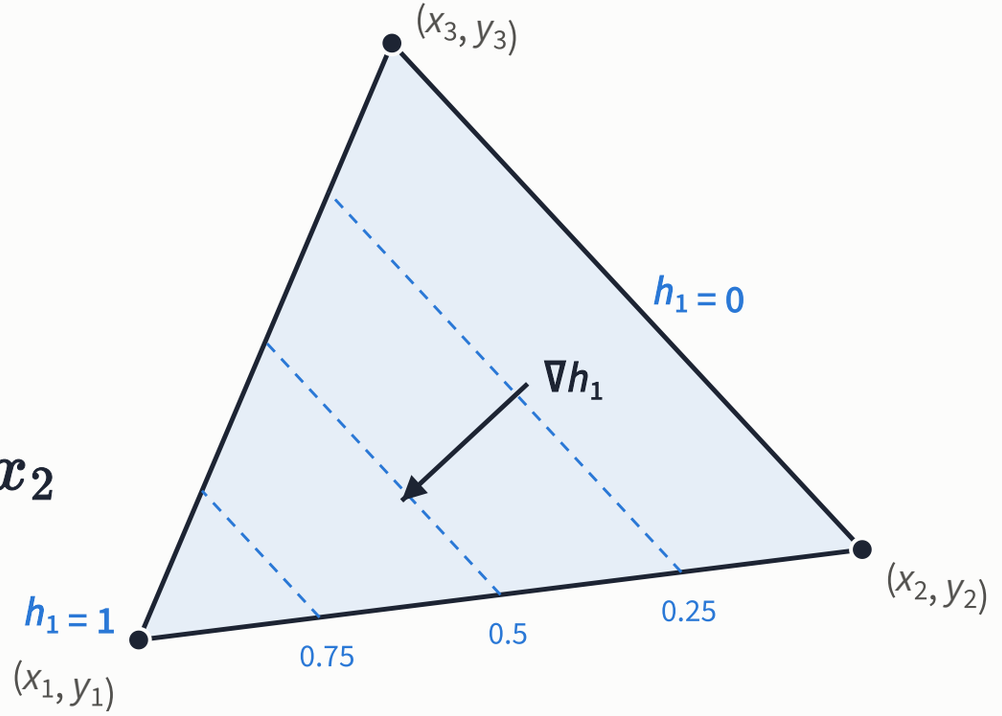
On a triangle with the vertices $(x_\alpha, y_\alpha), \alpha = 1, 2, 3$ (counterclockwise), u_h is linear and fixed by the three nodal values:

$$u_h = \sum_{\alpha=1}^{3} U_\alpha \, h_\alpha, \qquad h_\alpha(x, y) = \frac{a_\alpha + b_\alpha \, x + c_\alpha \, y}{2A}$$

$$a_1 = x_2 \, y_3 - x_3 \, y_2, \qquad b_1 = y_2 - y_3, \qquad c_1 = x_3 - x_2$$

and the other coefficients by cyclic permutation of the indices; A is the area of the triangle.

$h_\alpha(x_\beta, y_\beta) = \delta_{\alpha\beta}$ and $h_1 + h_2 + h_3 = 1$ (area coordinates): as in 1D, h_α is the hat function of the local node α, restricted to the triangle.



Element matrices of the CST

Constant gradients

$$\nabla h_\alpha = \frac{1}{2A}\begin{pmatrix} b_\alpha \\ c_\alpha \end{pmatrix}$$

constant on the triangle, hence the name **constant strain triangle**

Element stiffness matrix

$$k^e_{\alpha\beta} = \int_{T_e} \nabla h_\alpha \cdot \nabla h_\beta \, dx = \frac{b_\alpha\, b_\beta + c_\alpha\, c_\beta}{4A}$$

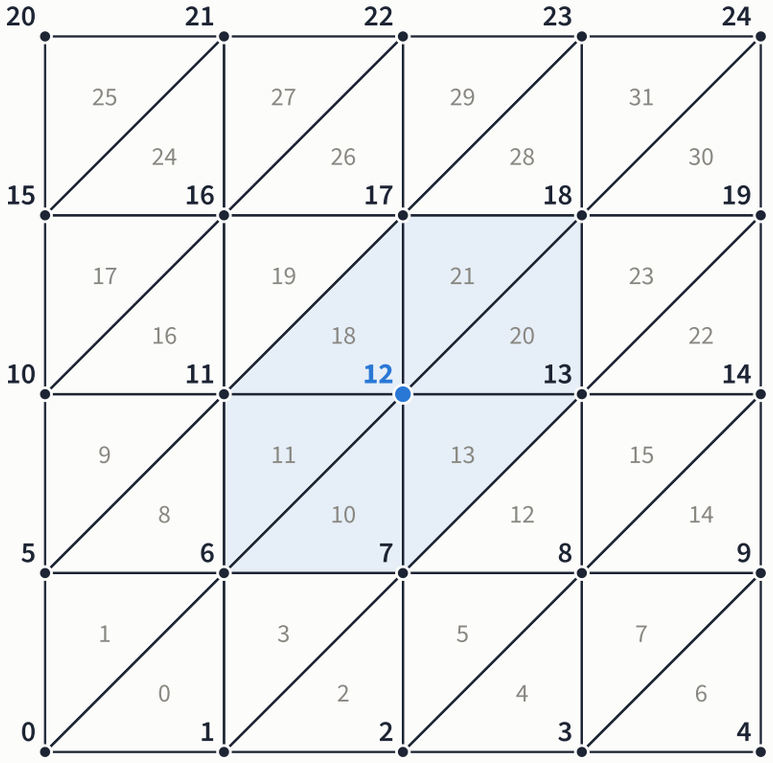
Element load vector, constant f

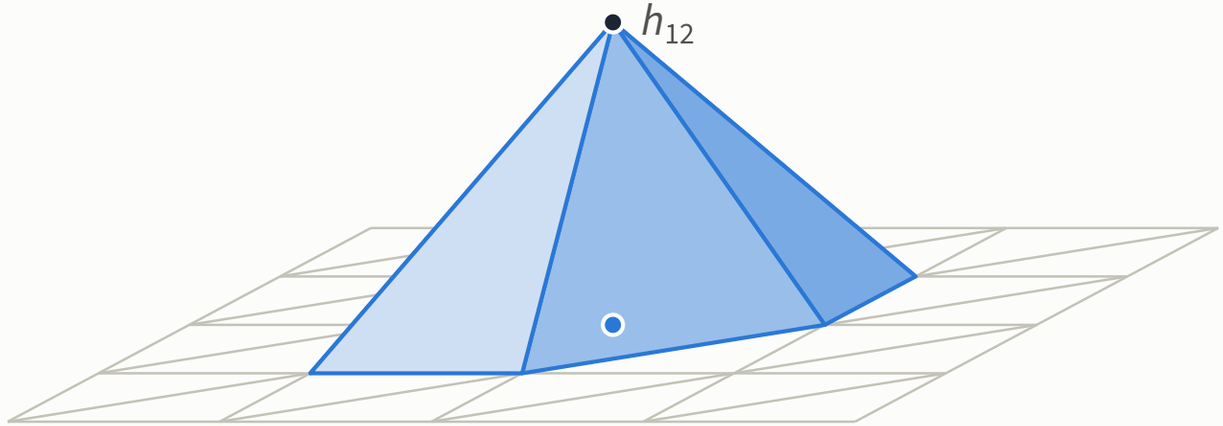
$$f^e_\alpha = \int_{T_e} f\, h_\alpha \, dx = \frac{fA}{3}$$

The global basis function h_i of node i is a **pyramid** of height 1 over the **patch** of triangles around node i.

Assembly works as in 1D, with the coincidence table of the mesh: three global node numbers per element, and k^e is 3×3.

A global basis function on a triangulation






```
mesh = RectangleMesh(Point(-1.0, -1.0), Point(1.0, 1.0), 4, 4)
V    = FunctionSpace(mesh, 'Lagrange', 1)
tri  = mesh.cells()           # coincidence table
v2d  = vertex_to_dof_map(V)   # node -> its basis function

h_i = Function(V)
h_i.vector()[v2d[12]] = 1.0
```

Node numbers black, element numbers grey: `tri[10]` = [6, 7, 12]. The patch of node 12 consists of the elements 10, 11, 13, 18, 20 and 21.

Element matrix, assembled matrix and solution

```python
u, v = TrialFunction(V), TestFunction(V)
a = inner(grad(u), grad(v))*dx
L = Constant(1.0)*v*dx

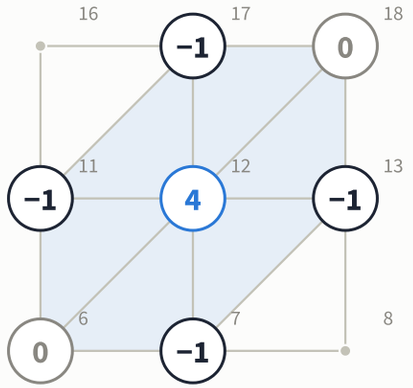
print(assemble_local(a, Cell(mesh, 0)))
K = assemble(a).array()
print(K[v2d[12], v2d])          # row of node 12

bc = DirichletBC(V, Constant(0.0), 'on_boundary')
uh = Function(V)
solve(a == L, uh, bc)
print(uh(0.0, 0.0))
```

Centre temperature $u_h(0,0) = 0.2813$, exact 0.2947: with only 9 unknowns, about 5 % too low.

Element 0, nodes 0, 1, 6:

$$k^e = \begin{pmatrix} 0.5 & -0.5 & 0 \\ -0.5 & 1 & -0.5 \\ 0 & -0.5 & 0.5 \end{pmatrix}$$



Row of node 12: 4 on the diagonal, -1 for the four neighbours: on right-angled triangles the CST gives the five-point finite difference stencil. The diagonal neighbours 6 and 18 get 0.

Sparsity

$K_{ij} = a(h_i, h_j)$ is nonzero only if the nodes i and j belong to a common triangle. A row of K has about seven nonzero entries, independent of the size of the mesh. For n unknowns:

	stored entries	memory for $n = 10^4$	memory for $n = 10^6$
dense matrix	n^2	0.8 GB	8 TB
sparse matrix (CSR)	$\approx 7n$	0.8 MB	84 MB

Direct solvers: fill-in

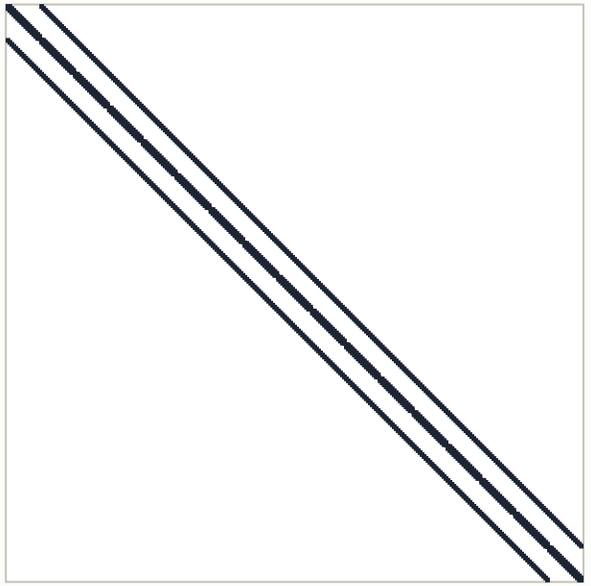
Gaussian elimination creates new nonzero entries in the triangular factors of K. Even with reordering, a sparse direct solver needs about $n^{1.5}$ operations in 2D and n^2 in 3D.

Iterative solvers

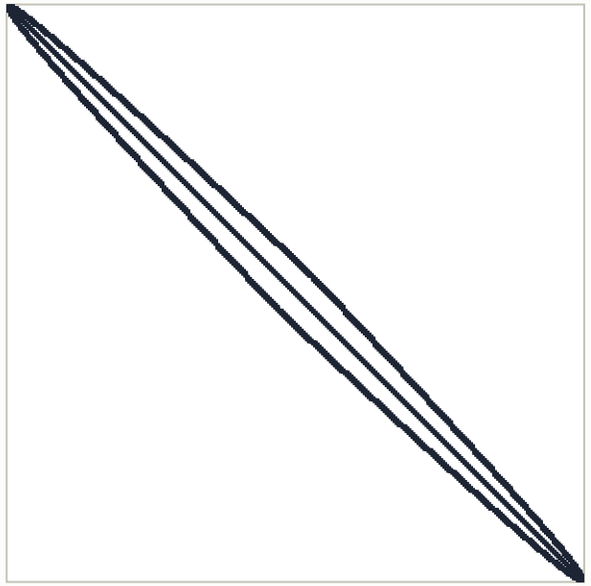
only need products $K\,p$: about $7n$ operations each, and K is never changed.

Sparsity pattern

node numbering



FEniCS numbering



The stiffness matrix of a 16×16 mesh, 289 nodes, in the order of the node numbers and in the order FEniCS uses. The numbering changes the pattern, not the number of entries.

n	nnz	nnz per row	n^2
289	1889	6.5	8.4e4
4225	29057	6.9	1.8e7
66049	460289	7.0	4.4e9
1050625	7346177	7.0	1.1e12

```
K = assemble(inner(grad(u), grad(v))*dx)
K.nnz()                    # stored entries
pyplot.spy(K.array())      # nonzero entries
```

nnz counts the stored entries, including the two zeros of the diagonal neighbours per row (example 3); the pattern shows only the nonzero ones.

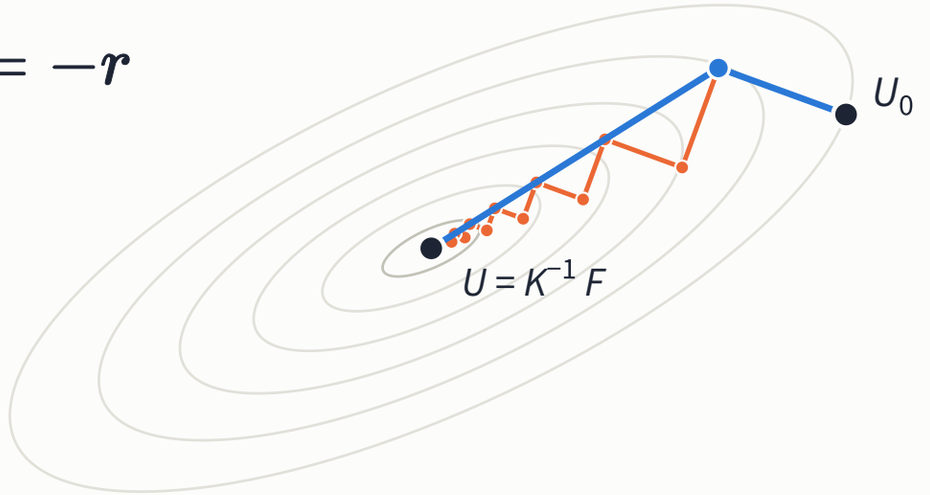
Solving $K\,U = F$ means minimizing

K is symmetric and positive definite. Solving $K\,U = F$ is then the same as minimizing

$$\Phi(U) = \tfrac{1}{2}\,U^T K\,U - U^T F, \qquad \nabla\Phi(U) = K\,U - F = -r$$

with the **residual** $r = F - K\,U$.

- **Steepest descent:** $U_{k+1} = U_k + \alpha_k\,r_k$ with $\alpha_k = r_k^T r_k / r_k^T K\,r_k$. It zigzags for an ill-conditioned matrix.
- **Conjugate gradient (CG) method:** the search directions p_k are K**-conjugate**, $p_j^T K\,p_k = 0$ for $j \neq k$. After k steps U_k minimizes Φ over $U_0 + \mathbf{span}\{r_0, Kr_0, \ldots, K^{k-1}r_0\}$ (Krylov space); the solution is reached after at most n steps.



The conjugate gradient method

$$
\begin{aligned}
&r_0 = F - K\,U_0, \quad p_0 = r_0 \\
&\text{for } k = 0, 1, 2, \ldots \\
&\qquad \alpha_k = r_k^T r_k \,/\, p_k^T K\,p_k \\
&\qquad U_{k+1} = U_k + \alpha_k\,p_k, \qquad r_{k+1} = r_k - \alpha_k\,K\,p_k \\
&\qquad \text{stop if } \|r_{k+1}\| \le \text{tol}\,\|F\| \\
&\qquad \beta_k = r_{k+1}^T r_{k+1} \,/\, r_k^T r_k, \qquad p_{k+1} = r_{k+1} + \beta_k\,p_k
\end{aligned}
$$

- Each iteration needs one product $K\,p_k$: about $7n$ operations.
- K is never changed, so there is no fill-in.
- The new direction p_{k+1} is the residual, made K-conjugate to p_k by adding $\beta_k\,p_k$.

Convergence and preconditioning

With the condition number $\kappa = \lambda_{\max}/\lambda_{\min}$ of K:

$$\|U - U_k\|_K \leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k \|U - U_0\|_K, \qquad \|e\|_K = \sqrt{e^T K\, e}$$

For element size h, $\kappa \sim h^{-2}$: the number of iterations grows like $1/h$, in 2D like $\sqrt{n}$, and the total work like $n^{1.5}$.

Preconditioning

Solve $M^{-1}K\,U = M^{-1}F$ with $M \approx K$ cheap to invert: the diagonal of K (Jacobi), incomplete Cholesky.

Algebraic multigrid (AMG)

The work is proportional to n: the number of iterations stays constant. In FEniCS:

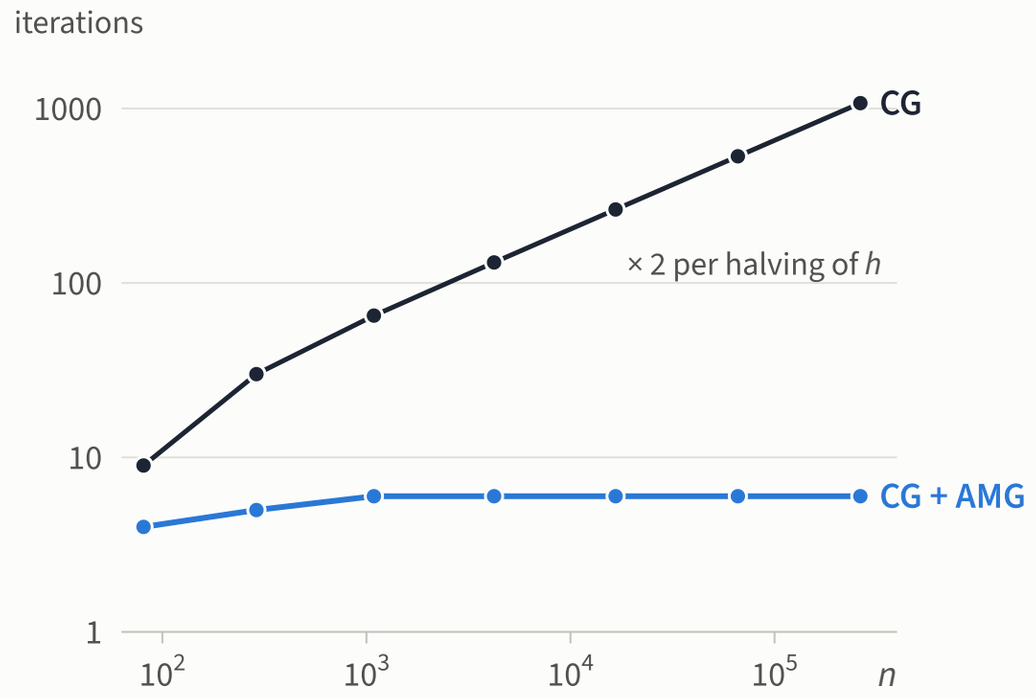
```
solver_parameters={'linear_solver': 'cg',
                   'preconditioner': 'amg'}
```

CG for the heated plate

```python
for N in [8, 16, 32, 64, 128, 256, 512]:
    mesh = RectangleMesh(Point(-1.0, -1.0), Point(1.0, 1.0), N, N)
    V    = FunctionSpace(mesh, 'Lagrange', 1)
    u, v = TrialFunction(V), TestFunction(V)
    bc   = DirichletBC(V, Constant(0.0), 'on_boundary')
    K, F = assemble_system(inner(grad(u), grad(v))*dx,
                           Constant(1.0)*v*dx, bc)

    for preconditioner in ['none', 'amg']:
        uh = Function(V)
        solver = KrylovSolver(K, 'cg', preconditioner)
        solver.parameters['relative_tolerance'] = 1e-10
        iterations = solver.solve(uh.vector(), F)
```

The residual is reduced by a factor 10^{10}. The error of the centre temperature drops from $3.6 \cdot 10^{-3}$ ($N = 8$) to $8.9 \cdot 10^{-7}$ ($N = 512$): like h^2.



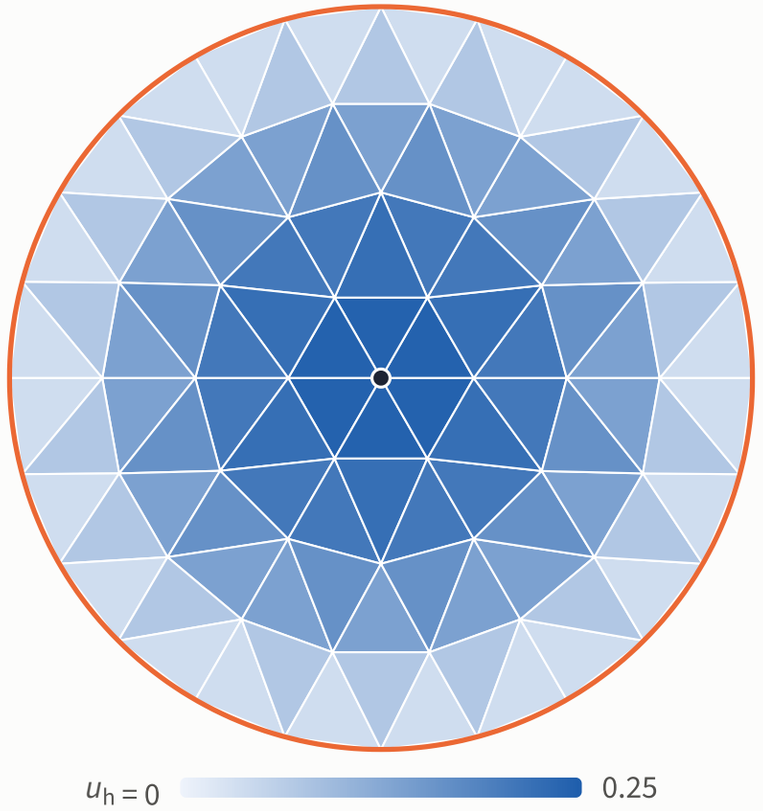
Without preconditioner the iterations double with every halving of h; with AMG they stay at about 6.

The mesh approximates the domain

Heated disk of radius 1, $-\nabla \cdot \nabla u = 1$, $u = 0$ on the boundary.
Exact: $u = (1 - r^2)/4$, centre temperature $1/4$.

```
mesh = UnitDiscMesh.create(MPI.comm_world, n, 1, 2)
V    = FunctionSpace(mesh, 'Lagrange', 1)
u, v = TrialFunction(V), TestFunction(V)
bc   = DirichletBC(V, Constant(0.0), 'on_boundary')
uh   = Function(V)
solve(inner(grad(u), grad(v))*dx == Constant(1.0)*v*dx,
      uh, bc, solver_parameters={'linear_solver': 'cg',
                                 'preconditioner': 'amg'})
```

h	0.620	0.337	0.175	0.089	0.045	0.023
n	19	61	217	817	3169	12481
error of $u_h(0,0)$	1.6e-3	7.3e-4	2.5e-4	8.0e-5	2.5e-5	6.7e-6



`UnitDiscMesh` replaces the boundary circle (orange) by a polygon; $n = 4$, 61 nodes. The error has two sources, the approximation of u and of Ω; both decrease like h^2.

The finite element method in four steps

1. Mesh the domain: nodes, elements T_e and the coincidence table (`mesh.cells()`).
2. One local basis function per node with $h_i(x_j) = \delta_{ij}$, so $U_i = u_h(x_i)$: hat functions in 1D, pyramids over the patch on triangles.
3. Compute the element matrices k^e, f^e and add them into K and F with the coincidence table. K is sparse, about 7 entries per row.
4. Solve $K\,U = F$ with CG. Without preconditioner the iterations grow like $1/h$; with AMG they stay constant: `{'linear_solver': 'cg', 'preconditioner': 'amg'}`.

EXERCISE 1

Conjugate gradients and preconditioners

1. Perform two steps of steepest descent and two steps of CG for

$$K = \begin{pmatrix} 3 & 2 \\ 2 & 6 \end{pmatrix}, \qquad F = \begin{pmatrix} 2 \\ -8 \end{pmatrix}, \qquad U_0 = (-2, -2).$$

   Check that CG reaches the solution $(2, -2)$ and that the two search directions are K-conjugate. Sketch the level sets of Φ and both paths.

2. Repeat example 5 with the preconditioners `'jacobi'` and `'ilu'`. How do the iterations grow with N? Why does Jacobi not change anything here? (Look at the diagonal of K in example 3.)

Linear elasticity: a dam

Find u with $u = u_D$ on Γ_D such that, for all v with $v = 0$ on Γ_D,

$$\int_\Omega \sigma(u) : \varepsilon(v)\, dx = \int_\Omega f \cdot v\, dx + \int_{\Gamma_N} T \cdot v\, ds,$$

$$\varepsilon(u) = \tfrac{1}{2}\left(\nabla u + \nabla u^T\right), \qquad \sigma(u) = \lambda\,(\operatorname{tr}\varepsilon)\, I + 2\mu\,\varepsilon.$$

Γ_D: displacement u_D prescribed; Γ_N: traction $\sigma n = T$ prescribed; f: body force.

Hydrostatic pressure of the water at the depth x: $g(x) = \rho\, g\, x$.

1. Which of $\Gamma_W, \Gamma_R, \Gamma_C, \Gamma_A$ are Dirichlet, which Neumann boundaries?
2. State the boundary condition on each part, with $g(x)$ on Γ_W, and modify the variational formulation accordingly.

